Concise Paper : Multi-Path TCP : Boosting Fairness in Cellular Networks

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***Abstract*—Cellular providers are rapidly deploying multiple technologies like cell biasing, carrier aggregation, co-ordinated interference control/scheduling to improve capacity and coverage. In this paper, we explore a complementary transport layer approach based on multipath TCP that can concurrently use multiple interfaces to boost throughput of users with poor coverage and improve fairness. Multipath TCP has been recently standardized by IETF and requires no modifications to applications. It has been shown to improve fairness and throughput in wireline environments and individual user throughputs in wireless networks. However, in a wireless multi-user environment, it is not clear that it is always beneficial, as we show in this paper. Therefore, we examine if it is indeed beneficial for a service provider to judiciously decide whether to enable multiple cellular interfaces on a smartphone based on a global centralized view of its network. Alternatively, should a device decide independently based only on a local view? To quantify the network wide impact in a system where users have multiple cellular interfaces, we have developed centralized and distributed heuristic algorithms to evaluate this, particularly in the context of fairness across all the users. Our simulations and numerical models show that there are potential gains in fairness (15-30%) to be realized by judiciously enabling multipath connections at the cell edge. These gains diminish as the number of users in a cell increases or users behave greedily. We also quantify the delicate balance between throughput and fairness. Our analysis provides an intuition on which user(s) in a cellular network stand to benefit the most by enabling multiple interfaces. We also discuss LTE protocol mechanisms to enforce associations of specific interfaces to specific cells.**

I. Introduction

Mobile data consumption is growing at an exponential rate. Users have an ever increasing appetite for more bandwidth and operators need to provide innovative solutions to keep up with bandwidth requirements and fairness over large coverage areas. Given the unique characteristics of a cellular network, this is particularly challenging for 'cell-edge' users who may suffer from weak signals and/or strong interference, thus receiving poor service.

There are several radio/MAC layer centric solutions being pursued to address this important issue such as cell biasing, inter-cell interference co-ordination (ICIC), co-ordinated multi-point scheduling (COMP), carrier aggregation *etc.* In this paper, we explore the possibility of improving cell-edge performance by devices that use multiple 'cellular' interfaces (on the same network) concurrently. Smartphones already carry multiple interfaces to talk to different networks *e.g.,* Wi-Fi and LTE. 3GPP is standardizing interfaces that can talk on several bands and we foresee devices that can communicate simultaneously on multiple cellular interfaces (laptops can currently do that by attaching multiple cellular cards).

Our study is devoted to investigating transport layer solutions (TCP), rather than the physical/MAC layer, that can leverage multiple interfaces. Multipath TCP (MPTCP) is a TCP extension being standardized in the IETF [1] that allows a single TCP connection to use multiple interfaces simultaneously. MPTCP works with unmodified applications, operates over today's internet and is designed to be fair to other users when contending for resources. The focus of this work is to investigate if and how we can leverage multipath TCP to improve coverage for 'edge' users, *i.e.,* improve 'fairness' of the cellular network. A key aspect of our approach is that we do not seek to modify any aspects of the underlying radio/MAC layer such as cellular scheduling, but instead improve performance by simply determining the number of interfaces a user should activate and which cell each interface should associate with. A clear benefit of this method is that it could be easily deployed with minimal change for user equipment (Ue), and simple 3GPP standardized procedures to set up connections between multiple interfaces and cells (see Section VI for related implementation aspects). Furthermore, it is complementary to existing radio/MAC layer approaches and could be overlayed upon them.

Although past work (*e.g.,* [5]) has looked at improving throughput and fairness in multi-user networks with MPTCP, they have focused primarily on wireline networks. Wireless networks however bring a different set of challenges. They have strong rate heterogeneity, with a user experiencing different rates from different cells unlike the largely homogeneous environment in wireline networks. In such an environment, when multiple users contend for scarce resources, it is unclear what benefits MPTCP yields, or if it could even be detrimental. Furthermore, activation of an additional wireless interface incurs penalties in the form of increased energy consumption for users, interference and reservation of signaling resources by the network. Hence any activation of an interface should be done judiciously and fully utilized to benefit the user. Finally, we wish to leave the underlying radio/MAC layer, especially the scheduler, undisturbed for ease of deployment. As discussed in Section II, this makes the problem complex due to limited degrees of freedom.

This paper presents some exploratory results to address these issues. We show with an analytic example the challenges of activating multiple interfaces in a heterogeneous rate environment. We then propose two contrasting classes of heuristics to leverage multipath TCP. The first class comprises of simple local heuristics wherein the users make independent

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decisions regarding activation/association of interfaces using local knowledge. The second class is 'centralized' heuristics, wherein a central processing node in the network (for example MME in LTE) receives signal strength information from all users and makes global decisions on which interfaces can be active and associate with which cells. Through numerical as well as detailed packet-based simulations that follow 3GPP standards and utilize the actual MPTCP kernel we evaluate the performance of our heuristics as well as potential benefits of MPTCP. The results show that with careful interface activation by centralized heuristics, MPTCP boosts fairness (by $15-30\%$ as measured by α-utility metric with $\alpha = 2$) as well as cell-edge (minimum) throughput (by $25 - 40\%$), while activating just a few additional interfaces (thus conserving resources). On the other hand, independent interface activation by users, provides small to no gains in fairness and can be quite inefficient. In addition, we also outline a 3GPP based protocol approach to implement our centralized heuristics that we plan to incorporate in a test-bed.

II. Analysis

A. Terminology and Assumptions

The system has K cells and N users that are assumed to be stationary. The MAC layer rate, r_{ij}, is the rate user i will receive from cell j if it were the only user on cell j. The radio channel experiences distance based path-loss. For simplicity, we ignore fast fading, though we plan to incorporate it in future work. Hence r_{ij} is only a function of distance d_{ij}. Our focus is on the downlink from cell to user. Each user initiates a continuous download so that buffers at all cells are assumed to be fully backlogged,*i.e.,* network is fully utilized (the case were some cells are under utilized would trivially result in straightforward improvements with multiple interfaces). We assume that cells use the **default** proportional fair scheduling to transmit to associated users. In a stationary channel with no fast-fading, it reduces to round-robin scheduling [11]. Consequently, if k users are associated with a cell, then user i on that cell gets an *effective* service rate of r_{ij}/k.

We assume that the throughput of a user attached to multiple cells is equal to the sum of its throughputs from individual cells. It is easy to modify the analysis in this section to the case where the throughput is a constant multiple of the sum of individual throughputs but we want to keep the exposition simple and use this section mainly to gain qualitative insight into the effect of a user making a second connection. We measure efficacy of our schemes with $\alpha-$fair metric defined as $F(\alpha) = \sum_{i=1}^{n} \frac{x_i^{1-\alpha}}{1-\alpha}$, where x_i is the total throughput achieved by user i. The $\alpha-$fair metric is a joint metric for fairness and throughput, parametrized by α, with larger values of α placing greater emphasis on fairness. E.g., $\alpha = 0$ corresponds to maximizing sum throughput, $\alpha = 1$ corresponds to proportional fairness, $\sum \log x_i$, and $\alpha \to \infty$ corresponds to the max-min fair metric.

B. Motivating example

We show by an example that indiscriminate second interface attachments can be detrimental. In particular, we illustrate a scenario where if every user made a second attachment (hoping to greedily increase its throughput), no user increases its throughout and many users see a reduction in their throughput. This underscores the need for the users to be judicious in making second attachments.

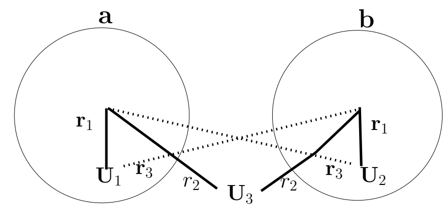


Fig. 1. Example showing that indiscriminate 2nd attachments result in loss of throughput

As shown in Fig. (1), assume two cells a and b and three groups of users: (1) U_1 is the set of n users at the center of the cell a, (2) U_2 is the set of n users at the center of the cell b, and (3) U_3 is the set of $2kn$ users on the edge of cells a and b.

Users in U_1 and U_2 receive rate r_1 from the cell close to them and rate r_3 from the far-off cell. Users in U_3 receive rate r_2 from both cells. By our assumption on proximity between groups and cells, we have $r_1 \gg r_2 \gg r_3$. Now we compare the throughput received by users under three attachment patterns.

(1) No multipath: The baseline is the scenario where each user attaches to exactly one cell. Users in U_1 attach to cell a; users in U_2 attach to cell b; and users in U_3 are equally split between a and b. (These attachments are shown by solid lines in Figure 1.) There are $n(1 + k)$ interfaces attached to each cell so the throughput of each user in U_1 and U_2 is $\frac{r_1}{n(1+k)}$ and throughput of each user in U_3 is $\frac{r_2}{n(1+k)}$.

(2) All users greedily attach to both cells: There are $2n(1+k)$ interfaces attached to each cell. For users in U_3, their new throughput, $\frac{r_2}{2n(1+k)} + \frac{r_2}{2n(1+k)} = \frac{r_2}{n(1+k)}$, is identical to their baseline throughput (in the scenario without any multipath).

For users in U_1 and U_2, their new throughput is $\frac{r_1}{2n(1+k)} + \frac{r_3}{2n(1+k)}$. The ratio of the allocated throughput to the baseline throughput is $\frac{1}{2} + \frac{r_3}{2r_1}$. If $r_3 \ll r_1$ then this ratio is close to $\frac{1}{2}$. *So, in conclusion, there is no throughput change for users in U_3 but close to 50% reduction for users in U_1 and U_2.* Intuitively a user gets resources proportional to its number of (attached) interfaces. By doubling the number of interfaces for each user, we do not change the fraction of resources users receive from the cells. This explains why users on the edge of the cells do not change their throughput. However, users near the center of the cell have significantly worse spectrum efficiency on their 2nd attachment (compared to their first attachment). They get the same amount of resources but do not have the ability to use those resources as efficiently as in the baseline.

(3) U_3 attaches to both cells; U_1 and U_2 maintain single attachment: There are $n(1 + 2k)$ users homed to each cell. Then throughput allocated to each user in U_1 or U_2 is $\frac{r_1}{n(1+2k)}$ and the ratio of the allocated throughput to the baseline throughput is $\frac{1+k}{1+2k}$.

For users in U_3, their allocated throughput is $\frac{r_2}{n(1+2k)} + \frac{r_2}{n(1+2k)}$ and the ratio is $\frac{2+2k}{1+2k}$. Table I shows these changes for various values of k. Overall we see a large increase in throughput for users in U_3 with small reduction for users in U_1 and U_2. We also observe that this pattern gets better if we have fewer users on cell edge compared to that in the

(Num of users on the cell edge) / (Num of users in the cell center)	%Dec in thpt of cell center users	%Inc in thpt of cell edge users
$1/2$	25	50
$1/4$	17	66
$1/20$	4	91

TABLE I. CHANGE IN THROUGHPUT IN THE 3RD SCENARIO

cell center. The intuition is that as we shift resources from cell center users to cell edge users, having fewer cell users translates into larger incremental resource per cell edge user.

III. HEURISTICS

In order to explore the benefits of MPTCP for fairness in a cellular environment, we propose and evaluate two classes of heuristics to control activation and attachment of a secondary interface : a) localized heuristics where users make local decisions based only on the information about their own rates and b) centralized heuristics where these decisions are made based on rate information from all users. Clearly, there can be intermediate distributed methods that allow a limited exchange of information, but our primary purpose is an initial exploration of the benefits of MPTCP in such an environment and the two extremal classes provide good reference points. For the former class, we propose two simple but practical heuristics, user-Random and user-Random-RSRQ.

A. Distributed: user-Random (β)

This very simple heuristic is defined by a parameter $\beta \in [0,1]$. *Every* user connects its primary interface to the cell with the best RSRQ[1] (which is the default behavior). It then initiates a second connection to the cell with the next-best RSRQ with probability β. Note that if $\beta = 0$, the scheme reduces to the default scheme (henceforth called $\mathbf{BaseLine}$), wherein every user connects to the cell with highest RSRQ.

B. Distributed: user-Random-RSRQ (β)

The user-Random-RSRQ heuristic builds upon user-Random by also taking into account the signal strength environment. Specifically, let for user i, $RSRQ_{i1}$ and $RSRQ_{i2}$ be the signal strength quality of the best and next-best cell. As before, the user will always attach its primary interface to the cell with the best RSRQ. Then, the user initiates a second connection to the cell with the 2nd best RSRQ with probability $\beta \cdot \frac{RSRQ_{i2}}{RSRQ_{i1}}$. The intuition is that the ratio measures the percentage gain in throughput the user will make by the second attachment, *if no other user were to make a 2nd attachment*. The example in Section II demonstrated that greedy attachment by all users to improve their throughput can deteriorate performance (case 1), but selective attachment by edge uses (case 3 : with similar throughput to both cells) can improve fairness. As such, this heuristic treats cell-edge users preferentially by allowing only users who can 'potentially' gain significant throughput from the second interface to connect. The factor β provides an additional parameter to modulate the likelihood of a secondary connection which can be tuned to explore behavior.

C. Centralized

As the name suggests, the Centralized heuristic seeks to decide multiple interface activation based on global information. 3GPP standards allow each user to report all its RSRQs to the associated cell, which in turn can report them to a central processing node (e.g MME). This central node can collect the RSRQs and build a *Rate Matrix* $[r]_{K\times N}$ where r_{ij} is the MAC layer rate that user i can expect from cell j (when only user in cell). It utilizes this information to decide on the optimal attachment for the primary interface, and if necessary, secondary interface of each user by using a heuristic outlined in Algorithm 1.

Algorithm 1 Centralized Greedy Heuristic

Require: Rate Matrix $\mathbf{R} = [\mathbf{r_{ij}}]_{\mathbf{N\cdot T\times K}}$, Utility Measure α, interfaces per user T.
Ensure: Optimal Attachment Matrix $\mathbf{A} = [\mathbf{a_{ij}}]_{\mathbf{N\cdot T\times K}}$, $a_{ij} \in \{0,1\}$.
Initial solution $\mathbf{A}$ = BaseLine.
$l = 1, LastChange = l, C = NUM(\mathbf{A}, \mathbf{R}, T, \alpha)$.
repeat $C' = \max_{\mathbf{e_l}} NUM(\mathbf{A}, \mathbf{R}, T, \alpha)$
  if $C' > C$ **then**
    $C = C'; LastChange = l;$
  end if
  $l = (l mod N \cdot T) + 1;$
until $l \neq LastChange$
function NUM($\mathbf{A}, \mathbf{R},, T, \alpha$)
  Compute $x_l = \sum_{j=1}^{K} r_{ij} \cdot a_{ij}/k_j \forall l = 1, \ldots N \cdot T$ where $k_j = \sum_{i=1}^{T} a_{ij}$.
  Compute $r_i = \sum_{i\cdot T+l=0}^{i\cdot T+l=T-1} x_l \forall i = 1, \ldots, N$. **return** $C = \sum_{i=1}^{N} \frac{r_i^{1-\alpha}}{(1-\alpha)}$.
end function

It is a greedy heuristic which iteratively goes through each of the $N \times T$ interfaces and tries to change the association for this interface while keeping all other associations unchanged.. There are $K+1$ possibilities for each interface: attach to one of the K cells or remain unattached. The heuristic iteratively steps through each interface and makes the choice that results in the largest increase of the α-utility and then moves to the next interface. It stops when no improvement is found after one full cycle (i.e all interfaces) from the last improvement.

IV. MULTIPLE INTERFACES : NUMERICAL EVALUATION

In this section, we explore several factors that can affect the benefits of multiple interfaces using a simple numerical model. In this model, K cells are placed equidistant on a circle of radius G with N users placed randomly within this circle. The MAC layer rate of user i from cell j is approximated as $r_{ij} = \log_2(1+SINR_{ij})$ where the received power is assumed to fade with distance d_{ij} as $1/d_{ij}^{3.5}$. Though simple, the model captures key aspects such as distribution of rates among users and allows us to quickly study and characterize various facets of multiple interfaces by reducing complexity via glossing over physical as well as transport layer packet dynamics.

In terms of multiple interface attachment, we consider possibilities along three dimensions: max number of cell that a user can attach to, subset of cell that a user can attach to (e.g., cell with highest RSRQs), and resource allocation among interfaces attached to a cell (Proportional fair or arbitrary ?). By examining the optimal for each possible trio, we get a sense of which of the dimensions are most critical. We also evaluate sensitivity w.r.t. number of users and choice of different values of α.

The numerical model also allows us to introduce, as a benchmark for evaluation, the 'optimal' association and resource allocation across multiple interfaces (hereby called $\mathbf{MultOpt}$). Specifically, MultOpt assumes that cell are not constrained by PF scheduling but can divide resources (time-slots) arbitrarily between attached users and that users can connect to any and all of the cell (i.e they can potentially

[1]RSRQ is the SINR of the pilot signal from a cell and hence proportional to the data rate that can be received from that cell. A user actively monitors RSRQ of all surrounding cells.

have $T = K$ interfaces). With these relaxations, the problem devolves to a simple convex optimization problem:

$$\max_{\theta_{ji}} \sum_{i=1}^{N} \frac{(\sum_{j=1}^{K} \theta_{ji} \cdot r_{ji})^{1-\alpha}}{1-\alpha} \quad , \tag{1}$$

$$s.t. \sum_{i=1}^{N} \theta_{ji} \leq 1, \quad \theta_{ij} \geq 0 \ \forall i, j, \tag{2}$$

where θ_{ji} represents the fraction of time-slots cell j assigns to user i so as to maximize $F(\alpha)$. MultOpt provides an upper bound on the maximum utility we can achieve.

A. Numerical Evaluation Results

For our evaluation, we assume 4 cells distributed on a circle of radius 500 m with a varying set of users placed randomly in the circle with locations generated uniformly. We set $\alpha = 2$ to emphasize fairness, but also evaluate for other α. For a specific setting of number of users, 50 instances were generated with varying locations. In each instance the algorithms were run to compute association and resource allocation and the fairness metric was computed from the resultant allocation.

Fig. 2(a) plots the CDF of the % change in the $\alpha = 2$ metric across *all* runs, of both Centralized, with $T = 1$ and $T = 2$ interfaces as well as MultOpt over the BaseLine for $N = 8$ users. (Recall that T is the maximum number of interfaces a user is allowed to have.) First, we observe that Centralized ($T = 2$) is quite close to MultOpt demonstrating its good performance. Second, we note that Centralized($T = 2$) provides significant gain ($\approx 30\%$) for $\alpha = 2$. The gain in fairness arises from two factors : **optimal association** and **multiple interfaces**. In the BaseLine scheme, each user associates with the highest RSRQ cell regardless of cell load. This can lead to poor load balancing among cells. By optimal selection of cells, a better load balancing can be achieved. Note that this does not require multiple interfaces and is addressed in past literature ([11]). The second gain is due to enabling multiple interfaces for a weak coverage (cell-edge) user to improve its throughput with slight reduction in throughput of a strong coverage (cell-center) user, resulting in overall improvement in fairness. The heuristics in Fig. 2(a) illustrate the impact of both features. Specifically, Centralized ($T = 1$) leverages only optimal association, whereas Centralized ($T = 2$) also leverages multiple interfaces. Interestingly, in scenarios where the user is allowed to have multiple interfaces, we found that in $\approx 85\%$ of the cases (both MultOpt and Centralized($T = 2$)), a single interface suffices, *i.e.,* only a few users need utilize multiple interfaces to boost network fairness.

Next, to explore the impact of user density, in Fig. 2(b), we look at the % gains as we increase the number of users to $20, 40, 80$ while keeping other parameters unchanged. A clear trend is observed in that the gain *decreases* with increasing number of users. Specifically, as the cell(s) become heavily loaded, the benefit of adding an additional interface decreases. We hypothesize that this is due to two reasons: a) given a uniform density, as the number of users increases, the cell load becomes homogeneous, resulting in fewer opportunities to load balance, and b) given the large number of users contending for a time-slot, activation of an additional interface yields a small time slice to the corresponding user, but takes away resources from a large number of other users.

Finally, we explore how multiple interfaces affects the α-utility metric. Note that a higher α value favors fairness over throughput. In Fig. 2(c) we plot % gain over BaseLine for $\alpha = 1, 2,$ and 3 with 8 users. As is clearly evident, the higher the emphasis on fairness (larger α), the more gains one obtains from multiple interfaces (*e.g.,* compare $\alpha = 3$ versus $\alpha = 1$). In other words, a network with more emphasis on fairness benefits from multiple interfaces which are turned on to boost the throughput of users with weak coverage (albeit at the expense of users with strong signal).

V. MPTCP: Simulation Results

We now build upon the analysis from Section IV that showed the benefits of multiple interfaces in a cellular environment by accounting for detailed MAC and TCP dynamics.

A. Simulation Framework

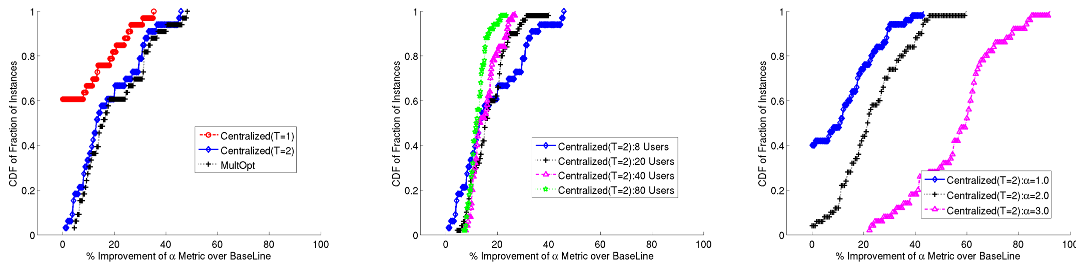
To account for network aspects such as packetization, MAC and transport layer characteristics we used the $ns-3$ [2] discrete event simulator (version 3.18) along with the LENA LTE module [3]. LENA incorporates a detailed model of the LTE downlink MAC layer that includes aspects such as signaling, handover, adaptive rate selection and opportunistic scheduling. Radio channel path-loss was modeled with the logdistance propagation model. We did not simulate fast fading losses on the channel, though we plan to incorporate it into future work.

For the transport layer, we utilized DCE (Direct Code Environment) [9] which allows $ns-3$ simulation code to directly use the Linux kernel stack. This allows us to use the *implemented* transport layer protocol code, thus closely accounting for real-world dynamics. The linux kernel chosen for our simulations incorporates MPTCP extensions [1] to enable us to exploit multiple interfaces.

B. Simulation Methodology

Similar to the setting of our numerical evaluation, we place 4 equidistant cells on a circle of radius 500 m and place users randomly within this circle. All cells transmit on the same channel and hence interfere uniformly across all subchannels. A common remote server opens a socket to each user and initiates a file transfer that runs for the duration of the simulation, which is 20 seconds. The application throughput is then measured at the end of the simulation. Each simulation was run 20 times with varying user locations. Note that since all users were continuously downloading, this results in a full backlogged situation with cell time-slot utilization close to 100% (unless a time-out causes starvation).

Each user's interface was associated with a cell using the 'Attach' and 'Handover' mechanisms. In the *localized* class of heuristics, a user can attach an interface to a cell by simply monitoring RSRQs and sending an 'Attach' request to the corresponding cell if they wish to activate an interface. For the centralized heuristics, each user activates a single (primary) interface, connects to the best RSRQ cell, and initiates the file transfer. It then reports observed RSRQs of all neighbor and reference cells via the standard 3GPP procedure implemented in LENA. All the RSRQs are collected by a central node that builds the RSRQ matrix and translates to a rate matrix based on RSRQ to MAC layer channel rates mapping. The controller then runs the centralized heuristic to determine the final optimal configuration. If the primary interface needs to associate with a different cell, a 'Handover' command is sent



(a) (b) (c)

Fig. 2. a) CDF of $\%$ increase in the $\alpha = 2$ fairness metric over BaseLine, b) Impact of users N ($\alpha = 2$), c) Impact of α ($N = 8$)

to the interface. If a second interface is to be used, we assume an 'Attach' command is sent.

We also simulated a modified version of this heuristic, Centralized-Aux, which runs the greedy heuristic *only* on secondary interfaces. The rationale here is that one wishes to minimize handover to avoid TCP time-outs, but also leverage multiple interfaces.

C. Simulation Results

We ran our simulations for 8 and 20 users uniformly distributed in a circle of 500m. The centralized heuristic optimized the association for $\alpha = 2$. Table II reports, for 8 users, the median $\%$ change in the α-utility metric across 20 runs for various heuristics compared to the BaseLine. For the localized heuristic we chose three values of $\beta = \{0, 0.5, 1.0\}$. Also, note that we report on changes for $\alpha = 0, 1, 2$ even though the Centralized heuristics optimize only for $\alpha = 2$, while the localized heuristics optimize local user throughput. By evaluating across all 3, we get a sense of the impact of multiple interfaces on metrics that weigh throughput ($\alpha = 0$) versus fairness ($\alpha = 2$).

We also include (last row), for purposes of comparison, results of the Centralized heuristic with a *Parallel* multi-path TCP controller. This is essentially, as the name suggests, two TCP sessions running independently on each interface, with the user getting the sum throughput. With a central controller, this scenario presents a useful approximation of the *upper bound* on the potential throughput one could obtain.

Focusing on the $\alpha = 2$ column, we note that *all* heuristics improve fairness with multiple interfaces, though to varying degrees. In particular, even with user-Random(1.0), where *every* user initiates two connections, we see improvement in fairness. While results from Section II indicate that users who greedily set up *independent* parallel connections can degrade fairness, a result we confirmed via simulations, two facets of the design and implementation of MPTCP prevents such a scenario. The first is coupled congestion control of all paths so as not to grab more resources than the fair share of TCP on the best path in a congested scenario, though in our experiments this did not play a primary role (especially if *all* users have multiple interfaces). A more dominant role was played by the packet scheduler implemented in MPTCP kernel which seeks to send data on subflows with the smallest RTT. In a cellular network, because of diversity, one link often has consistently higher throughput (lower RTT) than the other and we observed that its traffic share would strongly dominate.

To verify this, in Fig. (3), we plot the average ratio of traffic sent between the two paths for each of the heuristics. Observe that for the *localized* heuristics, user-Random, user-

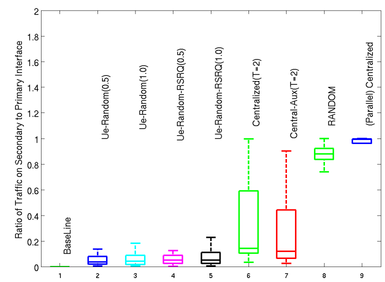


Fig. 3. Box-Plot of Ratio of Traffic on Secondary to Primary Interface for users with two connections

Random-RSRQ, this ratio is extremely small (≈ 0.1) indicating that the path with lower throughput is used sparingly. As a control experiment to verify our hypothesis, we modified the packet scheduler in the kernel to choose a path randomly with equal probability, while leaving the coupled congestion control untouched. The ratio of traffic for this scheme used in conjunction with user-Random (1.0) is plotted under the legend 'RANDOM' and is clearly much more evenly balanced with a ratio close to 1. Another interesting observation is that user-Random performs better than user-Random-RSRQ, though utilizing far more interfaces. We hypothesize that this is primarily because the latter allows fewer additional interfaces, but MPTCP does not fully utilize them.

Heuristic	% Gain over BaseLine					
	8 Users			20 Users		
	$\alpha = 2$	$\alpha = 1$	$\alpha = 0$	$\alpha = 2$	$\alpha = 1$	$\alpha = 0$
user-Random (0.5)	2.6	0.78	-0.05	12.93	2.46	-0.11
user-Random (1.0)	8.62	0.90	-0.25	10.57	0.30	-0.42
user-Random-RSRQ (0.5)	0.4	0.0	-0.01	11.68	2.74	0.00
user-Random-RSRQ (1.0)	4.87	1.59	-0.03	6.94	0.87	-0.18
Centralized	22.0	8.18	0.13	8.39	1.37	-0.06
Centralized-Aux	19.24	8.15	0.16	16.49	5.32	0.13
(Parallel)Centralized	29.24	12.63	0.01	35.79	9.99	-0.29

TABLE II. PERFORMANCE OF HEURISTICS

While the increase in fairness with *localized* heuristics that blindly deploy MPTCP is encouraging, Fig. (3) shows that this is achieved by MPTCP re-arranging flows to frequently use *one* (best) path. This is quite desirable in wireline networks, but in wireless networks, keeping interfaces active constantly, but not using them *fully* is inefficient due to energy usage for the user and signaling resources for the network. From Table II, the best performance for both localized heuristics come with $\beta = 1.0$, where all users activate both interfaces, though as we showed, typically one interface strongly dominates. In contrast, results from Section IV-A show only few users need activate interfaces to achieve the maximum utility. Consequently a scheme that *fully* utilizes a *few* active interfaces would be more desirable.

From that perspective we next look at the centralized heuristics. Table II shows that the centralized decisions result in a much larger gain ($19-20\%$ compared to 4%) due to better decisions on interface activation. In addition, from Table III, we see that they are more efficient than the optimum local heuristic, activating ($1.3 - 1.5$ interfaces per user, compared to 2). Another noteworthy aspect from Fig. (3) is that the centralized heuristics have a higher balance ratio indicating that active interfaces are better utilized.

Finally, as a comparison point, we note that the Centralized heuristics with the *parallel* MPTCP controller yield the largest gains with a noticeable gap from even centralized heuristics with the regular MPTCP controller. This is primarily because of the sparse usage of the secondary interface by MPTCP, an issue which we hope to address in future work.

Moving across the columns, we note that the Centralized heuristics improve $\alpha = 1$ as well as $\alpha = 0$ metrics, though the gains diminish rapidly. This is to be expected since a) the heuristics optimize for $\alpha = 2$, and b) multiple interfaces, as shown in Section IV-A, primarily benefit higher α values which emphasize fairness. The localized heuristics have negligible impact on these metrics.

While the α-utility metric summarizes *aggregate* network fairness and throughput, we are also interested in per-user throughputs. To that end, Table III shows, in the second and third columns, the average $\%$ gain in minimum and median throughput compared to the BaseLine scenario. Again, the Centralized heuristics provide large gains for minimum throughput, improving it by 37%, while improving median throughput by $\approx 8\%$.

Heuristic	8 Users			20 Users		
	Interfaces	% Thrpt Gain		Interfaces	% Thrpt Gain	
		Min.	Median		Min	Median
user-Random (0.5)	1.46	25.62	1.59	1.5	18.03	-3.2
user-Random (1.0)	2.0	32.93	-0.89	2.0	22.01	-8.28
user-Random-RSRQ (0.5)	1.29	14.76	3.27	1.27	17.66	-0.72
user-Random-RSRQ (1.0)	1.58	27.51	1.77	1.61	17.55	-3.79
Centralized	1.35	37.02	7.94	1.22	14.62	0.66
Centralized-Aux	1.48	37.01	7.35	1.35	20.52	2.72
(Parallel) Centralized	1.35	40.87	9.32	1.22	48.92	-2.47

TABLE III. INTERFACE AND THROUGHPUT STATISTICS

Moving next to the case with 20 users, we note some interesting differences. Firstly, the fairness gains are smaller as is to be expected based on observations in Section IV-A. Secondly, for $\alpha = 2$, the 'Centralized-Aux' heuristic performs best across *all* heuristics, while the 'Centralized' heuristic performance deteriorate. A closer inspection revealed that this is primarily because of handovers of the primary interface which degrades TCP throughput. Given the already smaller gains, the handoff penalty has a larger impact.

Finally, Table III shows that the heuristics boost the minimum throughput, with the overall best-performing Centralized-Aux heuristic boosting both minimum (20%) *and* median throughput. In summary, we observe that while MPTCP generally increases fairness, blind usage can be inefficient and also detrimental with increased load. By judicious activation, only a few interfaces are sufficient to leverages MPTCP.

VI. PROTOCOL TO CONNECT CELLULAR INTERFACES

We now discuss mechanisms in LTE that can be potentially used to attach/detach cellular interfaces from the perspective of a centralized heuristic. Current standards already require cells to periodically collect measurements from users which are forwarded regularly to a networking monitoring system (OSS). A central node can use this data to compute optimal associations. To enforce these associations, we envision usage of 3GPP standards based network-initiated handoffs. Specifically, a controller can use an API (called FAPI interface) to instruct the source cell to handover to a target cell. For interfaces which need to be de-activated, the MME can initiate a detach procedure by sending a DETACH REQUEST message with detach TYPE 're-attach not required'. We are currently exploring implementation of these procedures on a testbed.

VII. RELATED WORK

Most of the existing work regarding multipath transport has focused on the *wired* context [4, 5, 10]. Recently, Chen et al. [7] have conducted measurements using MPTCP over Wifi and cellular networks, albeit from the context of throughput of a single user. There is a large body of literature, e.g., [6, 8, 11] to name a few, that addresses the issue of both single and multiple interface association for multiple users in a wireless network. However, they presume that the cell (base-station) resource allocation is flexible and can be modified based on the desired optimal solution. Further, they focus only on MAC layer performance. Our framework leaves the cell untouched, instead trying to achieve fairness in a multi-user wireless network by manipulating the number of connections a user can have, which is more practical. More importantly, we also explore transport layer dynamics by determining how MPTCP can be leveraged to exploit multiple interfaces.

VIII. CONCLUSION

We have presented several exploratory results via numerical modeling and detailed simulations that analyze MPTCP for improving cellular coverage. The key observation we make is that MPTCP is not always beneficial in the wireless context. Efficiency and fairness gains can diminish if multiple interfaces are not activated judiciously. To that extent, we propose centralized decision making algorithms that can be used to improve the fairness greatly by deciding when to activate multiple interfaces on an individual user basis. Our current on-going work is to (a) improve the MPTCP packet scheduler, (b) develop robust distributed heuristics, and (c) implement a centralized controller to enforce attach/detach in an LTE testbed.

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